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CHEMICAL REACTION ENGINEERING (CHE 321)

QUIZ 1

CLOSED BOOK, TIME 15 MIN

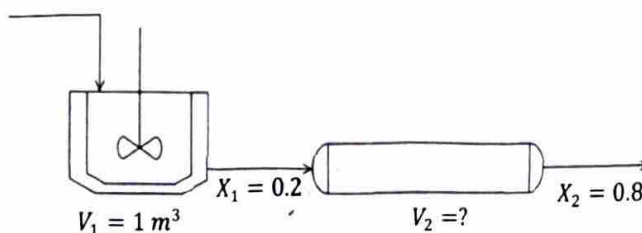
Consider the liquid-phase reaction $A \rightarrow B$, with the rate as a function of conversion given as

$$r_A = -kC_{A0}(1 - X)$$

, where k is a rate constant, C_{A0} is the concentration of species A in the feed.

The reaction is to be carried out isothermally in a continuous stirred tank reactor (CSTR), followed by a plug flow reactor (PFR). Volume and conversion of the CSTR are 1 m^3 and 20%.

Calculate the volume of the PFR required to achieve 80% final conversion.



Hint: The General mass balance equation is $\frac{dN_A}{dt} = F_{A,in} - F_{A,out} + G_A$

Mole Balance for CSTR

$$F_{A0} - F_A + r_A V = 0$$

$$F_{A0} = C_{A0} V_0$$

$$V = \frac{F_{A0} - F_A}{-r_A} \quad F_A = F_{A0} - F_{A0} X_1$$

$$V = \frac{F_{A0} - F_{A0} + F_{A0} X_1}{-r_A} = V_1 = \frac{F_{A0} X_1}{-r_A}$$

$$V_1 = 1 \text{ m}^3 \quad X_1 = 0.2$$

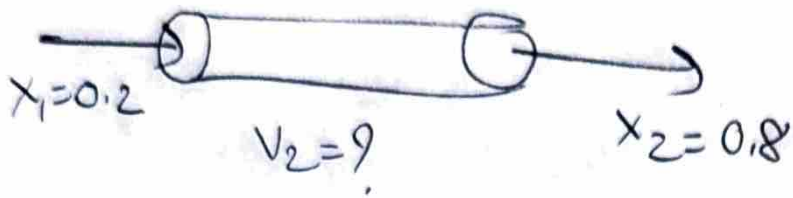
$$\frac{1 \text{ m}^3}{0.2} = \frac{F_{A0}}{-r_A} = \boxed{5 \text{ m}^3}$$

Since this
is a CSTR
 $r_A = -kC_A(1-X)$

$$r_{A1} = -kC_{A0}(1 - X_1)$$

$$r_{A1} = -kC_{A0}(1 - 0.2)$$

$$r_{A1} = -kC_{A0}(0.8)$$



$$V_2 = \frac{\Delta x}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + f_4]$$

$$V_2 = \left(\frac{F_{A0}}{-r_A} \right) \int_{0.2}^{0.8} dx$$

$$\Delta x = \frac{0.8 - 0.2}{2} = 0.3$$

$$F_{A0} - F_A = \Delta V r_A$$

$$r_A = K (A_0) (1-x)$$

$$r_A = K (C_A)$$